

BengaliAddressNet: A Weighted Soft-Voting Ensemble for Bengali Handwritten Upazila–District Pair Name Recognition

Abstract—Recognizing handwritten Bengali addresses is challenging due to the script’s cursive nature, diverse writing styles, and high visual similarity among upazila-district names. Prior work has mostly focused on isolated characters or words, leaving multi-word address pairs largely unexplored due to the lack of a dedicated dataset. We address this gap with a novel dataset (BNHW-UDPNR) of 9,808 handwritten upazila-district pair images from 320 pages, segmented using an unsupervised MSER and Canny edge-based pipeline. For recognition, we propose BengaliAddressNet, an ensemble of Vision Transformer (ViT-Base) and ConvNeXt-Tiny combined via weighted soft voting, achieving 94.65% test accuracy, a macro-F1 of 0.94, and a ROC-AUC of 99.99%.

Index Terms—Bengali handwritten recognition; Upazila–district pair; Vision transformer; ConvNeXt; Ensemble learning; Text segmentation

I. INTRODUCTION

Transforming handwritten documents into a digital format is a primary objective in computer vision and image processing. In Bangladesh, many routine administrative tasks, such as sorting mail, registering land, student enrollment, and collecting census data, are still conducted on paper. In these handwritten documents, including mail, land records, and government forms, the names of upazila and district often appear next to each other. Bangladesh has 8 divisions, 64 districts, and 496 upazilas. Identifying these handwritten location names is important for automating the sorting and indexing process of documents.

Bengali ranks as the seventh most widely spoken language in the world, with over 230 million people using it as their native language [1]. Although it is commonly used, research on Bengali handwriting recognition has not been as extensively explored as that on Western scripts. The Bengali script provides unique challenges owing to its design. It consists of an intricate set of fundamental characters, vowel modifications (matras), linked letters (consonant conjuncts), and a unique horizontal line (shirorekha) that connects characters into words [2]. Handwritten Bengali in real life poses extra challenges due to its varied writing styles, pen thickness variations, tilted writing lines (skew), and background noise or ink blots [3], [4].

Most current systems focus only on identifying individual characters or words [5], [6]. The automatic reading of full, multi-word address lines, such as a combined upazila-district pair, is still unexplored. This task is challenging because many

upazilas across various districts look very similar or share the same spellings, creating visual confusion.

In this study, we have created: (1) a new labeled dataset (BNHW-UDPNR) containing 9,808 upazila-district pair images; (2) an automated unsupervised text segmentation pipeline; and (3) BengaliAddressNet, an ensemble method that incorporates a Vision Transformer [7] (ViT-Base) with a convolutional network ConvNeXt-Tiny [8] for upazila-district pair handwritten images recognition.

II. RELATED WORK

A. Deep Learning Approaches

Deep neural networks have significantly advanced feature extraction for complex scripts. Chakraborty et al. [9] introduced a Deep CNN (DCNN) that achieved 93.07% character recognition accuracy, while Hamid et al. [10] developed a modified ResNet-34 architecture tailored for precise character recognition. Validated on a merged subset of popular handwriting datasets, this deep convolutional framework achieved state-of-the-art results, reaching an overall accuracy of 97.00% on mixed text and peaking at 99.02% for numeric characters. To address the low recognition rates of complex compound characters and modifiers, Islam et al. [11] introduced RATNet. Evaluated against benchmarks like VGG-16 and ResNet-50, this framework successfully reduced parameter complexity while achieving peak multi-class character recognition accuracies ranging from 97.70% to 99.66% on the CMATERdb dataset. At the word level, Zisad et al. [12] designed a CNN-based classifier optimized for handwritten Bengali district names, reaching a 98.44% test accuracy.

Chowdhury et al. [13] introduced a Vision Transformer (ViT) for holistic cursive script recognition. The architecture modeled global contextual dependencies from sequential patches, showing a performance increase of up to 26% after incorporating data augmentation.

To address character alignment issues and visual script similarities, Rabby et al. [5] introduced BornoNet, a simple and lightweight Convolutional Neural Network (CNN) engineered for the classification of 50 basic isolated Bengali characters. Evaluated across CMATERdb, ISI, and BanglaLekha-Isolated, the model achieved peak validation accuracy of up to 98

Parsing geographical data within address fields introduces substantial intra-class variation. Addressing this challenge, Tusher et al. [14] investigated handwritten Bangladeshi city name variations by compiling a comprehensive geographic

script dataset, benchmarking performance across deep learning networks (e.g., ResNet, DenseNet, and Inception) to build a specialized model for automated document routing.

B. Machine Learning Approaches

Before deep learning, handwritten Bengali recognition relied primarily on hand-engineered descriptors. Mollah et al. [15] proposed a holistic recognition method utilizing a pixel density histogram descriptor that captures structural geometry across elliptical, tetragonal, and vertical spaces. When evaluated on a custom public database of 18,000 Bengali word images over 120 categories, their system achieved a peak recognition accuracy of 83.64% using a Support Vector Machine (SVM) classifier.

Nazmul Hoq et al. [16] presented a comparative overview of seven traditional classification algorithms for Bengali Handwritten Character Recognition (BHCR) evaluated across multi-dimensional performance metrics. Testing on a newly compiled dataset of 2,500 handwritten vowel samples preprocessed via histogram adjustment, their empirical analysis demonstrated that Artificial Neural Networks (ANNs) yielded the highest overall accuracy, while Logistic Regression proved exceptional in optimizing structural sensitivity and specificity thresholds.

C. OCR Systems and End-to-End Text Extraction

Complete address digitization requires integrating robust object localization with text extraction. Maung et al. [6] established a scalable handwritten Bengali OCR pipeline combining YOLO for character detection, EfficientNet-B4 for recognition, and a Word2Vec spelling correction layer to minimize character error rates. Similarly, Mazumder et al. [17] developed an end-to-end automated Bengali signboard recognition system optimized for resource-constrained edge environments. Their framework achieved a peak signboard localization accuracy of 99.10% on a massive natural-scene dataset.

Shakib et al. [18] developed a deep learning framework for joint character localization and classification. Combined with a dedicated post-processing text-assembly routine, their intelligent OCR pipeline achieved a character classification accuracy of 91.00% and an overall document text-extraction accuracy of 86.59%.

To overcome the architectural complexities of subword tokenization in dense scripts, Hasan et al. [19] developed GraDeT-HTR, a resource-efficient Bengali Handwritten Text Recognition system utilizing a specialized grapheme-aware decoder-only transformer. Pretrained on large-scale synthetic datasets and fine-tuned on real human-annotated samples, their model captures intricate character conjuncts and diacritics directly, establishing a state-of-the-art performance baseline across multiple public text-line benchmarks.

Most previous studies focus on recognizing single characters or single words [5], [6], [9]–[12], [15], [16], with little work on combined fields like upazila-district pairs [14], largely due to the lack of a dedicated public dataset. Existing methods

also use convolutional [9]–[12], [15]–[17] or transformer [13] models separately, and rely on supervised detection requiring manual labeling [6], [17], [18]. To address these gaps, this paper proposes BengaliAddressNet, a ViT-Base and ConvNeXt-Tiny ensemble with weighted soft voting, supported by a new dataset (BNHW-UDPNR) and an unsupervised segmentation pipeline requiring no manual annotation.

III. RESEARCH METHODOLOGY

The Bengali upazila–district pair recognition framework uses a four-stage process: creating the dataset, segmenting text within images, preprocessing matrices, and classifying sequences with an ensemble approach.

A. Dataset Collection

The BNHW-UDPNR dataset includes handwriting samples for all 496 Bengali upazila-district pairs. Writers of different ages contributed, creating natural variation in style, size, and angle. The dataset comprises 320 pages: 60% from students, 40% from teachers, and the remaining 0.5% from other professionals. High-definition cameras captured the handwritten sheets as RGB images. To add variety, postal envelopes and school or college admission forms were included. The dataset is available on Kaggle [20].

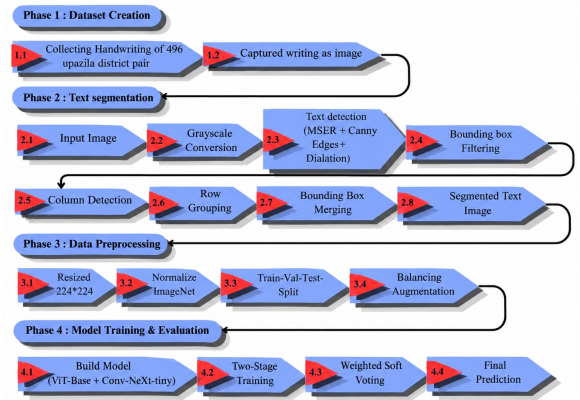


Fig. 1. Overall working process of Bengali Upazila–District name recognition.

B. Text Segmentation Pipeline

An unsupervised text segmentation pipeline was developed with two phases: preprocessing and segmentation. Preprocessing converts RGB images to grayscale, applies Otsu’s thresholding for binarization, reduces salt-and-pepper noise, and uses a 9×3 dilation kernel to connect text components horizontally.

Two methods were used to detect text before identifying candidate regions. The first is MSER Detection with $\delta = 5$, $\text{min_area} = 150$, and $\text{max_area} = 10,000$. The second uses Canny Edge Detection with thresholds of 50 and 150, followed by two rounds of 9×3 dilation. Bounding boxes from both methods are filtered by size, keeping those 10–900 pixels wide and 10–250 pixels high.

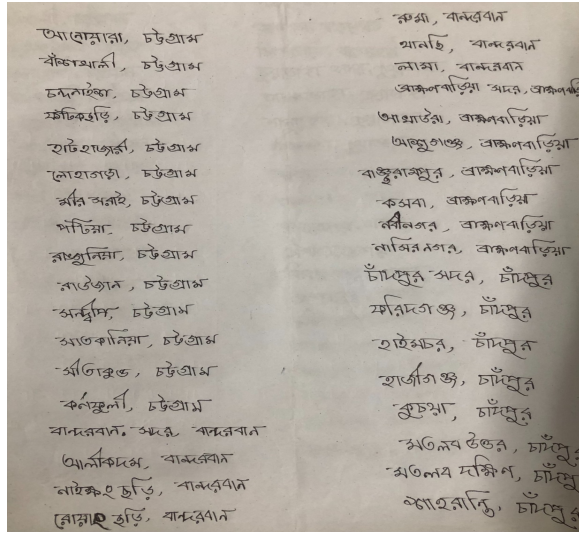


Fig. 2. Handwritten sample grayscaled

Connected Component Analysis (CCA) is applied with 8-connectivity to detect diagonal lines common in Bengali compound characters. Elements smaller than 50 pixels are removed as noise, resulting in initial bounding boxes at the character, word, or line level.

Column splitting uses a vertical projection profile to separate left and right regions. In each column, boxes are grouped into rows if they are within 90 pixels of each other vertically. Boxes are merged if they are next to each other horizontally (gap of 35 pixels or less) or overlap vertically by at least 45% of the smaller box's height. OpenCV's minAreaRect finds a rotated bounding rectangle for each group, and an affine transformation corrects skew before cropping. About 90% of images were segmented automatically, while 10% needed manual refinements.

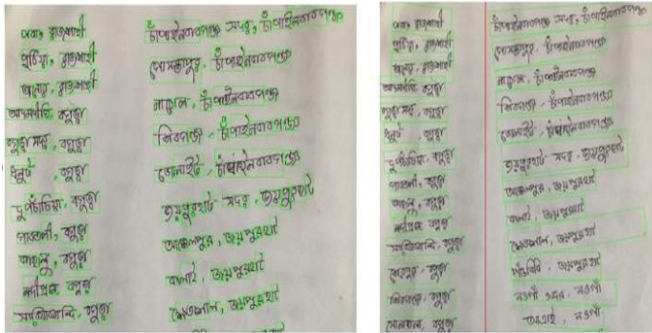


Fig. 3. (a) Connected component analysis with Bounding Boxes. (b) Text-line bounding boxes across a dual-column layout

C. Data Preprocessing and Augmentation

The segmented images are scaled to 224x224 pixels through bilinear interpolation and adjusted using ImageNet statistics (mean [0.485, 0.456, 0.406]; std [0.229, 0.224, 0.225]). The dataset is divided into training (70%), validation (20%), and

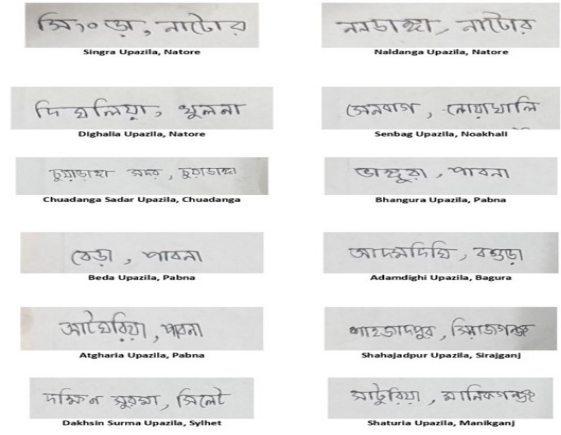


Fig. 4. Segmented images

test (10%) sets using a stratified split, with classes containing fewer than 5 samples removed.

To address class imbalance, a balancing augmentation approach increases the underrepresented classes to match the largest class by randomly combining geometric (affine, elastic) and photometric (brightness, noise, blur) transformations with document-specific (ink dropout, ink bleed, uneven light) transformations. Deduplication uses difference hashing (dHash) to avoid the creation of unnecessary synthetic images. In training, augmentations additionally contain random rotation ($\pm 5^\circ$), random affine (3% translation, 5° shear), random erasing (0.25 probability), and Mixup [21] ($\alpha = 0.1, 0.50$ probability starting from epoch 2).

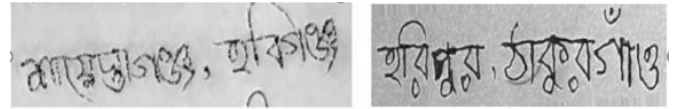


Fig. 5. Data augmented samples: (a) Uneven illumination and (b) elastic deformation.

Prior to augmentation, the training, validation, and test sets contained 6,828, 1,953, and 1,047 images, respectively, across 496 classes. After applying the balancing augmentation strategy described above, the training set was expanded to 11,904 images, while the validation and test sets remained unchanged at 1,949 and 1,047 images, respectively, guaranteeing that evaluation was performed exclusively on unaugmented data.

IV. PROPOSED MODEL ARCHITECTURE

The proposed ensemble model, BengaliAddressNet, uses a Vision Transformer (ViT-Base) and a ConvNeXt-Tiny. Both models connect to the same MLP classification head, which produces logits for 496 classes. Their outputs are combined using soft-weighted voting.

BengaliAddressNet has two parallel branches: a Vision Transformer (ViT-Base) and a ConvNeXt-Tiny model. Both link to a custom MLP head. Their logits, before softmax,

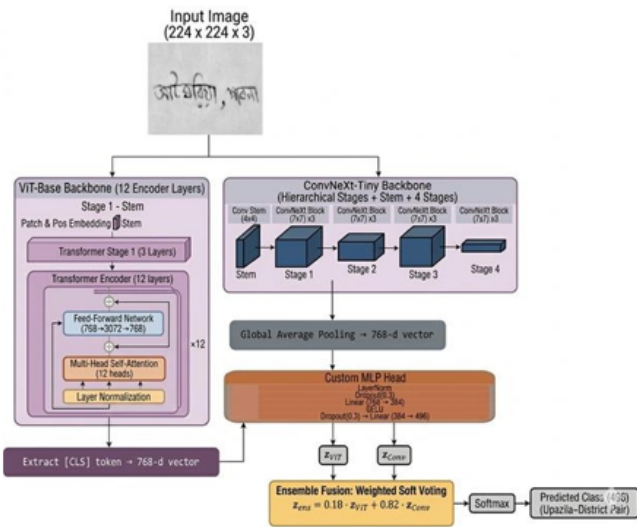


Fig. 6. Architecture of BengaliAddressNet: ViT-Base and ConvNeXt-Tiny branches with shared custom MLP head, combined via weighted soft voting ($w=0.18$ for ViT, 0.82 for ConvNeXt).

are combined using weighted soft voting. The next sections explain each branch and the shared classification head.

A. Vision Transformer (ViT-Base) Branch

ViT-Base handles a $224 \times 224 \times 3$ input by dividing it into 196 distinct 16×16 -pixel patches. Each patch is converted into a 768-dimensional vector by a trained linear transformation. Learnable positional embeddings and a distinct [CLS] token are added before the 197-token sequence is passed through 12 transformer encoder layers, each containing 12-head multi-head self-attention (MSA) and a $768 \rightarrow 3072 \rightarrow 768$ MLP block using GELU activation. The output from the CLS token is used as input into the custom classification head.

B. ConvNeXt-Tiny Branch

The ConvNeXt-Tiny model starts with a 4×4 section that moves with a stride of 4. Then a four-part block (3,3,9,3) starts to learn. Each of these four blocks has 96, 192, 384 and 768 channels, respectively. Every block has a 7×7 depth wise convolution, layer normalization, 1×1 convolution (expanding channels 4 times), GELU activation, and a projection in order to return in original size. Global average pooling creates a 768-dimensional feature vector, which is then used as input to the custom head.

C. Custom Classification Head

The custom MLP head gets two inputs, one from the ViT-Base branch and another one from the ConvNeXt-Tiny branch.

The custom MLP head receives a 768-dimensional feature vector from either branch, the [CLS] token output from ViT-Base or the global average pooled output from ConvNeXt-Tiny and maps it to 496 output logits corresponding to the 496 unique upazila–district classes. This head architecture is applied independently to both branches with separate learned

weights, enforcing a consistent projection structure while allowing each branch to adapt its representations to the task.

The custom MLP head processes the 768-dimensional input through six sequential steps: (1) Layer Normalization stabilizes activations across the feature space; (2) Dropout(0.3) prevents co-adaptation and reduces overfitting; (3) Linear($768 \rightarrow 384$) compresses the representation to its most discriminative features; (4) GELU activation preserves non-zero gradients for negative inputs, improving gradient flow over ReLU, as confirmed in Table II; (5) a second Dropout(0.3) adds further regularization; and (6) Linear($384 \rightarrow 496$) projects to 496 output logits, one per upazila–district class, consumed directly by cross-entropy loss during training and softmax during inference.

D. Weighted Soft Voting Ensemble

Weighted soft voting is an ensemble fusion strategy in which each classifier’s predicted class probabilities are combined as a weighted linear sum to produce a final decision [22]. The proposed model, BengaliAddressNet, combines the pre-softmax logits of both branches through this strategy, preserving the full confidence information from each model.

Let Z_{ViT} and $Z_{Conv} \in \mathbb{R}^{496}$ denote the logit vectors from ViT-Base and ConvNeXt-Tiny respectively. The ensemble logits are computed as:

$$Z_{ens} = \omega \cdot Z_{ViT} + (1 - \omega) \cdot Z_{Conv} \quad (1)$$

The predicted class is $\hat{y} = \arg \max_k \text{softmax}(Z_{ens}[k])$. The weight ω is optimized by exhaustive grid search over $[0, 1]$ in steps of 0.01 on the validation set.

V. EXPERIMENTS AND RESULTS

A. Experimental Setup

All models were trained on Google Colab using an NVIDIA Tesla T4 GPU (16 GB VRAM) with PyTorch 2.0. The BNHW-UDPNR dataset uses a 70/20/10 train/validation/test split. Performance is evaluated using accuracy, macro-averaged F1-score (Macro-F1), weighted-average F1-score (Weighted-F1), and ROC-AUC (one-vs-rest strategy). The best validation checkpoint is retained for final testing.

B. Model Performance Comparison

Fig. 7 provides a grouped bar chart visualizing macro-averaged recall and F1-score margins across all baseline models and the soft-voting framework. The graphical distribution confirms that the soft-voting ensemble reliably bridges performance drops found in isolated classifiers.

C. Model Training Curve Analysis

The learning behavior of the individual branches throughout full backpropagation is evaluated using structural training metrics.

As illustrated in the training and validation accuracy and loss tracking graphs, both neural pathways exhibit smooth, stable asymptotic convergence without manifesting signs of gradient destabilization or overfitting.

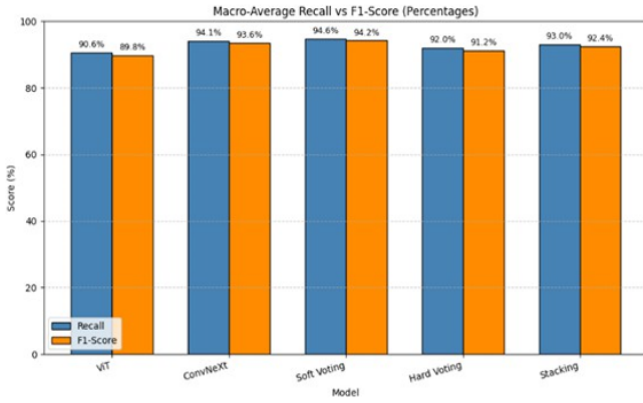


Fig. 7. Grouped bar chart of recall and F1-score (macro) for each model

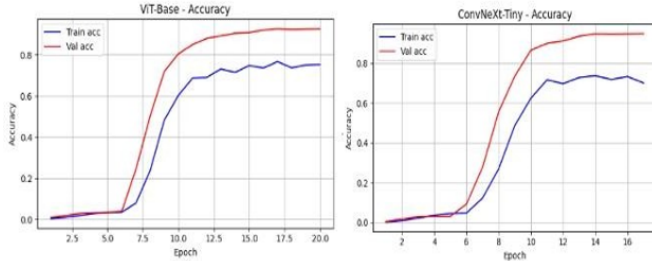


Fig. 8. Training and validation accuracy for ViT-Base and ConvNeXt-Tiny.

D. Results and Discussion

As shown in Table I, ConvNeXt-Tiny outperforms ViT-Base individually (94.17% vs. 91.12% test accuracy), confirming that convolutional inductive biases are particularly beneficial for handwritten Bengali text where local stroke patterns are critical. The ensemble method we used, BengaliAddressNet, further improves upon ConvNeXt-Tiny by +0.48% test accuracy, demonstrating the value of combining complementary architectures.

The overall test accuracy of 94.65% with a macro-F1 of 0.94 and ROC-AUC of 99.99% demonstrates strong discriminative capability across all 496 administrative classes. Misclassification analysis reveals that errors predominantly occur between upazilas within the same district (e.g., Daganbhuiyan vs. Chagalnaiya, both in Feni district), reflecting script similarity

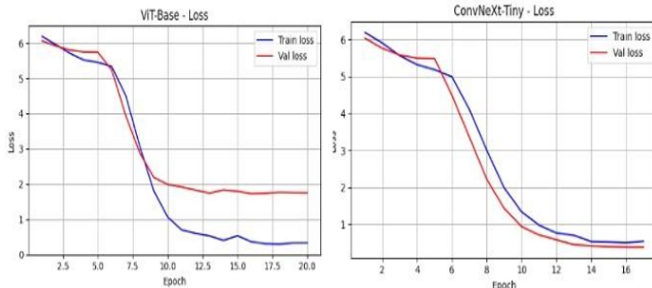


Fig. 9. Training and validation loss for ViT-Base and ConvNeXt-Tiny.

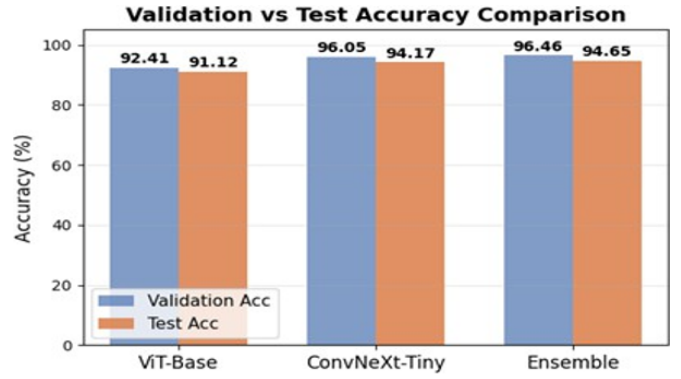


Fig. 10. Validation vs. test accuracy comparison for ViT-Base, ConvNeXt-Tiny, and weighted soft voting ensemble (BengaliAddressNet).

rather than model weakness. The precision-biased behavior (low false-positive rate) is desirable for practical postal routing applications.

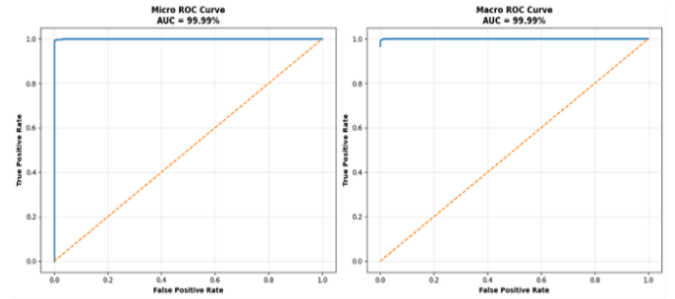


Fig. 11. ROC-AUC curves (micro and macro average) for the BengaliAddressNet on the test set (AUC = 99.99%).

E. Effects of Activation Function (ReLU vs. GELU)

TABLE I
EFFECT OF ACTIVATION FUNCTION (ReLU vs. GELU) ON TEST ACCURACY

Model	Activation Functions	
	ReLU Test Acc (%)	GELU Test Acc (%)
ViT-Base	89.54	91.12
ConvNeXt-Tiny	93.28	94.17

Table I confirms that GELU activation consistently outperforms ReLU for both architectures, with a more pronounced improvement for ViT-Base (+1.54%) than ConvNeXt-Tiny (+1.13%). This aligns with established literature showing that smooth, non-zero negative activations better support self-attention feature representations.

F. Effect of Data Augmentation

To quantify the contribution of the balancing augmentation strategy, all four model configurations were retrained and evaluated under identical hyperparameter settings but with *no data augmentation* applied to the training set. Preprocessing

(resizing to 224×224 and ImageNet normalization) was retained; only the geometric, photometric, document-specific, and Mixup augmentations described in Section III-C were removed. Table II reports the resulting accuracy, precision, recall, macro-F1, and weighted-F1 scores on the test set.

TABLE II
EFFECT MODEL PERFORMANCE WITHOUT DATA AUGMENTATION

Model	Classification Metrics				
	Acc. (%)	Prec. (%)	Rec. (%)	Macro-F1. (%)	Wtd-F1 (%)
Vit-Base	79.18	80.04	78.26	76.54	77.48
ConvNeXt-Tiny	87.97	87.66	87.11	85.74	86.69
BengaliAddressNet	91.21	91.30	90.89	89.83	90.22

Comparing Table II against Fig. 10 reveals consistent and substantial improvements across all models when augmentation is applied. The weighted soft voting ensemble gains the largest absolute improvement: from 91.21% to 94.65% test accuracy (+3.44 percentage points). Individual models benefit even more markedly: ViT-Base improves by +11.94 percentage points (79.18% → 91.12%) and ConvNeXt-Tiny by +6.20 percentage points (87.97% → 94.17%). The macro-F1 score of the ensemble rises from 89.83% to 94.18%, indicating that augmentation especially benefits underrepresented classes rather than only the majority classes.

VI. CONCLUSION

This paper presented an end-to-end system for recognizing handwritten Bengali upazila–district pairs, addressing a critical gap in low-resource language document processing. A novel dataset (BNHW-UDPNR) of 9,808 images covering all 496 administrative pairs was constructed from 320 diverse writers. An unsupervised segmentation pipeline combining MSER detection, Canny edge detection, column analysis, and minAreaRect bounding box merging reliably extracts individual text regions. BengaliAddressNet, a weighted soft voting ensemble of ViT-Base and ConvNeXt-Tiny with optimal weights [0.18, 0.82], achieves 94.65% test accuracy, with a macro-F1 of 0.94 and ROC-AUC of 99.99%, outperforming individual models and alternative ensemble strategies.

Future work includes dataset expansion for underrepresented classes, deep learning-based segmentation to reduce the 10% manual adjustment rate, model compression for edge deployment, and extension to full address recognition incorporating house numbers, road names, and postcodes.

REFERENCES

- [1] “Bengali language,” Wikipedia, 2026, accessed: Jun. 24, 2026. [Online]. Available: https://en.wikipedia.org/w/index.php?title=Bengali_language&oldid=1360606801
- [2] U. Pal and B. B. Chaudhuri, “Indian script character recognition: a survey,” *Pattern Recognition*, vol. 37, no. 9, pp. 1887–1899, Sep. 2004.
- [3] M. Z. Alom, P. Sidike, M. Hasan, T. M. Taha, and V. K. Asari, “Handwritten bangla character recognition using the state-of-the-art deep convolutional neural networks,” *Computational Intelligence and Neuroscience*, vol. 2018, pp. 1–13, Aug. 2018.
- [4] S. Bag and G. Harit, “A survey on optical character recognition for bangla and devanagari scripts,” *Sadhana*, vol. 38, no. 1, pp. 133–168, Feb. 2013.

- [5] A. S. A. Rabby, S. Haque, S. Islam, S. Abujar, and S. A. Hossain, “BornoNet: Bangla handwritten characters recognition using convolutional neural network,” *Procedia Computer Science*, vol. 143, pp. 528–535, 2018.
- [6] A. T. Maung, S. Salekin, and M. A. Haque, “A hybrid approach to bangla handwritten OCR: combining YOLO and an advanced CNN,” *Discover Artificial Intelligence*, vol. 5, no. 1, p. 119, Jun. 2025.
- [7] A. Dosovitskiy *et al.*, “An image is worth 16x16 words: Transformers for image recognition at scale,” 2020, arXiv.
- [8] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, “A ConvNet for the 2020s,” 2022, arXiv.
- [9] P. Chakraborty, A. Islam, M. A. Yousuf, R. Agarwal, and T. Choudhury, “Bangla handwritten character recognition using convolutional neural network,” in *Machine Intelligence and Data Science Applications*, ser. Lecture Notes on Data Engineering and Communications Technologies, V. Skala, T. P. Singh, T. Choudhury, R. Tomar, and M. A. Bashar, Eds., vol. 132. Singapore: Springer Nature Singapore, 2022, pp. 721–731.
- [10] M. A. R. Alif, “State-of-the-art bangla handwritten character recognition using a modified resnet-34 architecture,” Jan. 2024.
- [11] M. S. Islam, M. M. Rahman, M. H. Rahman, M. W. Rivolta, and M. Ak-taruzzaman, “RATNet: A deep learning model for bengali handwritten characters recognition,” *Multimedia Tools and Applications*, vol. 81, no. 8, pp. 10631–10651, Mar. 2022.
- [12] M. H. Zisad, S. M. Ayman, A. Chowdhury, M. I. Siddiquee, and M. Hasan, “BN-HW-DSNd: A dataset of handwritten bengali words for text recognition,” in *Proceedings of the 3rd International Conference on Computing Advancements*. Dhaka, Bangladesh: ACM, Oct. 2024, pp. 862–867.
- [13] A. Chowdhury, M. A. Hossen, M. Hasan, and S. M. R. Osmani, “Performance analysis of vision transformer based architecture for cursive handwritten text recognition,” in *2023 26th International Conference on Computer and Information Technology (ICCIT)*. Cox’s Bazar, Bangladesh: IEEE, Dec. 2023, pp. 1–6.
- [14] M. M. R. Tusher *et al.*, “Development of a lightweight model for handwritten dataset recognition: Bangladeshi city names in bangla script,” *Computers, Materials & Continua*, vol. 80, no. 2, pp. 2633–2656, 2024.
- [15] S. Bhowmik, S. Malakar, R. Sarkar, S. Basu, M. Kundu, and M. Nasipuri, “Off-line bangla handwritten word recognition: a holistic approach,” *Neural Computing and Applications*, vol. 31, no. 10, pp. 5783–5798, Oct. 2019.
- [16] M. N. Hoq, N. A. Nipa, M. M. Islam, and S. Shahriar, “Bangla handwritten character recognition: an overview of the state of the art classification algorithm with new dataset,” in *2019 1st International Conference on Advances in Science, Engineering and Robotics Technology (ICASERT)*. Dhaka, Bangladesh: IEEE, May 2019, pp. 1–6.
- [17] T. Mazumder, F. Nusrat, A. B. S. Mahi, J. B. Brishty, R. Rahman, and T. Helaly, “Automated and efficient bangla signboard detection, text extraction, and novel categorization method for underrepresented languages in smart cities,” *Results in Engineering*, vol. 26, p. 105156, Jun. 2025.
- [18] W. Shakib, M. I. Khalil, A. T. M. A. B. Quader, S. M. R. Osmani, M. Hasan, and A. Chowdhury, “Deep learning based text extraction system from handwritten bengali word,” in *2023 26th International Conference on Computer and Information Technology (ICCIT)*. Cox’s Bazar, Bangladesh: IEEE, Dec. 2023, pp. 1–6.
- [19] M. M. Hasan, A. N. T. Choudhury, M. Hasan, and M. M. Khan, “GraDeT-HTR: A resource-efficient bengali handwritten text recognition system utilizing grapheme-based tokenizer and decoder-only transformer,” in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*. Suzhou, China: Association for Computational Linguistics, 2025, pp. 696–706.
- [20] D. M. Islam, “upazila-district-final-v2,” Kaggle, 2024, accessed: Jun. 26, 2026. [Online]. Available: <https://www.kaggle.com/datasets/d22232mojahidulislam/upazila-district-final-v2>
- [21] H. Zhang, M. Cisse, Y. N. Dauphin, and D. Lopez-Paz, “mixup: Beyond empirical risk minimization,” 2017, arXiv.
- [22] G. Fumera and F. Roli, “A theoretical and experimental analysis of linear combiners for multiple classifier systems,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 27, no. 6, pp. 942–956, Jun. 2005.